Student Worksheet: Solving Linear Equations

IGCSE Mathematics | Brackets, fractions and real-world equations

Name: _____ **Date:** _____

Learning goals

- Expand brackets correctly.
- Keep equations balanced by doing the same operation to both sides.
- Clear fractions using a common multiple.
- Check solutions by substitution.

Remember: SBIC

S — Simplify	Expand brackets or clear fractions first.
B — Balance	Perform the same operation on both sides.
I — Isolate x	Get the variable by itself.
C — Check	Substitute your answer into the original equation.

A. Starter: Taxi challenge

A taxi charges a ¥10 starting fee plus ¥4 for every kilometre. The total fare is ¥34.

1. How many kilometres did the taxi travel? Show your mental or written method.

Working: _____

2. Write an equation using x for the number of kilometres: _____

3. What does x represent? _____

B. Worked example: brackets

Study the steps, then complete the check.

$3(x + 2) = 18$

$3x + 6 = 18$

$3x = 12$

$x = 4$

Check by substitution: 3(_____ + 2) = 3 × _____ = _____

Why must the 3 multiply both terms inside the brackets?

_____

C. Guided practice: negative brackets

Solve: $5 - 2(x + 1) = 3x - 2$. Show each step.

_____

_____

_____

_____

_____

Check: substitute your value of x into the original equation. Do both sides match? _____

D. Misconception check

A student says: $4(x - 3) = 20$ becomes $4x - 3 = 20$.

1. What mistake has the student made?

_____

2. Write the correct expanded equation: _____

E. Fractions: clear the denominators

Solve: $(x + 3)/3 + x/6 = 4$

1. The denominators are 3 and 6. Their lowest common multiple is: _____

2. Multiply every term by the LCM. Write the new equation:

_____

3. Solve the equation, showing your steps:

_____

_____

_____

_____

4. Why must you multiply the 4 on the right-hand side too?

_____

F. Challenge: fractions, brackets and subtraction

Solve: $(2x - 1)/3 - (x + 2)/4 = 5$

Hint: the LCM of 3 and 4 is 12. Multiply every term by 12.

_____

_____

_____

_____

_____

_____

G. Real-world application: gaming café

You and a friend each play for 3 hours and each buy a ¥6 drink. Together, you pay ¥60.

1. Let x be the cost per hour of gaming. Write an equation: _____

2. Solve for x, showing your working:

_____

_____

_____

3. The cost per hour is ¥_____.

H. Exit ticket — independent

Solve: $3(2x - 1) + 5 = 20$. Show every step.

_____

_____

_____

_____

_____

My confidence with SBIC today (circle one): 1 2 3 4 5

One step I need to practise more: _____